

EMMANUEL OLADELE OKUNLOLA

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RESEARCH PROFILE

Research-focused engineer with strong foundations in robotics, control systems, and machine learning for autonomous systems. My work centers on developing safe, intelligent, and resilient cyber-physical systems, with applications in autonomous vehicles and robotic systems. I am particularly interested in bridging theoretical control models with real-world deployment in resource-constrained environments.

EDUCATION

- **Ekiti State University** Ado-Ekiti, Nigeria
Bachelor of Technology Education in Mechanical and Auto Mechanical Technology Sep 2017 – Jun 2023
(B.Tech.Ed.), Second Class Upper Division (**CGPA: 4.36/5.00**)
 - Relevant coursework: Introduction to Mechatronics and Robotics, Basic Electricity, Mechanical Drawing, Numerical Methods, Mathematics and Basic Calculus, Auto Workshop Management, Mechanical Drawing, Internal Combustion Engines (ICE) and Fuel Systems.
 - **Research interests:** Control of autonomous vehicles (aerial and terrestrial), Cybersecurity and resilience of cyber-physical systems, Machine Learning for Robotics, Simultaneous Localization and Mapping (SLAM)

RESEARCH AND TEACHING EXPERIENCE

- **Robotics and Artificial Intelligence Nigeria (R.A.I.N)** Ibadan, Nigeria
Course Instructor and Lead Researcher Jan 2026 – Present
 - Delivered hands-on instruction in robotics and intelligent systems, covering ROS2, path planning, digital twin technologies, and embedded systems development.
 - Taught cloud deployment for web applications using tools like Render and AWS, enabling students to deploy AI and robotics solutions to scalable production environments.
 - Mentored and supervised student research projects across robotics, AI, and autonomous systems, guiding problem formulation, system design, implementation, and evaluation.
 - Supported the development of robotic systems by integrating sensors, microcontrollers, and control algorithms for real-world applications.
 - Facilitated project-based learning by assisting students in debugging, testing, and optimizing their solutions.
- **Anglican Commercial Grammar School** Ibadan, Nigeria
Student Teacher Sep 2019 – Dec 2019
 - Taught and graded Basic Technology for 60 Junior Secondary School students in Junior Secondary Schools 1 and 2.
- **Baptist Community High School** Oyo, Nigeria
Student Teacher Sep 2021 – Dec 2021
 - Taught and graded Technical Drawing for 24 Senior Secondary School students in SSS 1 and SSS 2.
- **Nesto College** Oyo, Nigeria
Instructor Jul 2022 – Dec 2022
 - Instructed Basic Technology for a class of 26 students in Junior Secondary Schools 1, 2, and 3.
 - Taught Technical Drawing for a class of 10 students in Senior Secondary Schools 1, 2, and 3.

PROFESSIONAL EXPERIENCE

- **Lendsqr** Remote, Nigeria
Product Specialist (Lender Operations) Oct 2024 – Dec 2025
 - Implemented ML and data science solutions to improve lender operations and loan processing efficiency.
 - Built a Credit Custom Model to estimate loan repayment probability, contributing to lower non-performing loans.
 - Automated ticketing workflows by integrating n8n with Google Chat for summarization and escalation, reducing median resolution time for escalated issues from approximately 2 hours to about 45 minutes.
 - Developed a multi-modal model that combines transaction narration, entry type, and date features to classify bank transactions and derive borrower debt-to-income metrics.
 - Designed structured intern training modules focused on customer support handling and escalation pathways.
- **One Campus Data Analysis (Cohort 17)** Remote, USA
Data Analyst Intern Jul 2022 – Jun 2024
 - Built a grocery-store manager prototype using Python (OOP) and Pandas to simulate inventory and sales logic.
 - Performed data cleaning, outlier handling, and exploratory analysis with Pandas and NumPy to support predictive tasks.
- **Freelance Python Developer** Remote
Software Developer Nov 2022 – Oct 2023
 - **Maze Searcher App:** Implemented maze rendering and solver algorithms (BFS, DFS) with Python Turtle and PIL; focused on algorithm correctness and visualization.
 - **Data Visualization with Processing:** Processed a global CO₂ emissions dataset (1751–2014) and produced interactive visualizations using Processing and GeoPy for geocoding.

- **Auto Clinic**

Automobile Mechanic Assistant - Industrial Training

- Assisted in vehicle diagnostics, maintenance, and road testing; validated repair outcomes and recorded results for quality control.
- Used CRM tools to manage vehicle delivery schedules and customer notifications.

Ibadan, Nigeria

Sep 2019 – Dec 2019

SELECTED PROJECTS

- **Multi-modal Transaction Classifier** — *Prototype model that uses narration text, entry type, and date features to classify bank transactions for credit evaluation.*
- **Electric Go-Kart (Research)** — *Designed and fabricated a microcontroller-based electric go-kart with gyroscope-assisted wireless steering input for improved vehicle stability and control.*
- **Mini Cannon Launcher (Research)** — *Built a compressed-air projectile launcher to study projectile dynamics and system safety considerations.*

SKILLS AND TOOLS

Programming and Data: Python, SQL, C++, Bash, L^AT_EX.

Data Science and ML: Pandas, NumPy, Scikit-learn, TensorFlow and Keras, PyTorch.

Systems and DevOps: n8n, Git, Docker, REST APIs, AWS, Render.

Robotics and Simulation: ROS2, Turtle, PyBullet,.

Embedded and Hardware: ESP8266, ESP32, Arduino, Raspberry Pi.

Tools and Software: MySQL, Fast Api, OBS, PowerPoint, MATLAB, Google Spreadsheets

VOLUNTEER EXPERIENCE

- **The Triumphant Baptist Church**

Multimedia Production Lead

Ibadan, Nigeria

Jan 2022 – Jun 2024

- Used video mixers, sound mixers, and camcorders with broadcasting software (OBS) to live stream church events to retain membership of the church during the lockdown period as a result of the COVID-19 pandemic, to maintain community engagement.
- Developed and implemented a Standard Operating Procedure (SOP) that established clear, stringent requirements for team membership and outlined structured guidelines for effective collaboration with other departments.
- To enhance team management and overall productivity, I established sub-teams, allocated responsibilities, and assigned experienced members as mentors to guide new trainees.
- Assess and capture videos to guarantee digital integrity to get the best video output, raising online membership of the church by 5% using software like Adobe Audition and Premiere Pro.

- **Speaking Engagements**

Guest Speaker and Volunteer Educator

Nigeria

2022 – Present

- Assisted in delivering a lecture to secondary school students at All Saints Church School, Ibadan, titled “*Emerging Careers in AI and Artificial Intelligence*”, introducing pathways in technology and innovation.
- Volunteered as a speaker at a Up-skilling conference themed “*Taking Flight*”, where I presented a lecture on “*Effective Prompt Engineering*”, focusing on practical applications of AI systems in day-to-day life.
- Contributed to the establishment of a robotics club for secondary school students; taught SCRATCH programming and introductory Python to build foundational computational thinking skills.

CERTIFICATIONS

- **Intermediate Python** — (2021)
- **Robotics and Automation Developer** — (2023)
- **Introduction to Generative AI AWS** — (2025)
- **Supervised Learning with scikit-learn** — (2025)
- **Meta AI Agents and Agentic AI Systems development** — (2025)